really is the only way to get where you're trying to go, but more likely, you've made a wrong turn somewhere a few miles back.

In practice, though, it doesn't even take something as clearly dysfunctional as a true singularity to lay waste to physics as we know it. Any time you have a lot of energy in a very small space, you have to deal with both quantum mechanics (the theory governing particle physics) and general relativity (the theory of gravity) at the same time. Under normal circumstances, you're only dealing with one or the other, because when gravity is important, it's usually because you have a massive thing, so you can ignore the individual particles, and when quantum mechanics is important, at the particle scale, you're dealing with so little mass that gravity is a totally negligible part of the interaction. But at extreme densities you have to contend with both, and they don't work well together, *at all*. Extreme gravity involves well-defined massive objects that warp space and alter the flow of time; quantum mechanics allows particles to pass through solid walls or exist only as fuzzy probability clouds. The fundamental incompatibility of our theories of the very massive and the very small is one of the things that hints at the direction we should go in creating new, more complete theories. But it is also rather inconvenient when we're trying to explain the very early universe.

Without a full theory of quantum gravity (something that reconciles particle physics with gravity), there's a limit to how far back we can extrapolate the universe in a way that makes sense. We inevitably reach a moment before which all bets are off. During that time, the densities are high enough that we expect extreme gravitational effects to be competing with the inherent fuzziness of quantum mechanics, and we just don't know what to do in that scenario. Do microscopic black holes form (because of the strong gravity) but then pop in and out of existence at random (because of the quantum uncertainty)? Does time have any meaning when the shape of space is no more predictable than a roll of dice? If you zoom in to a small enough scale, do space and time act like discrete particles, or perhaps waves that interfere with each other? Are there wormholes? Are there dragons??? We have no idea.

But because we need to quantify exactly how confused we are, and at what moment that confusion sets in, we call this the Planck Time,[fn13] and it encompasses the time from zero to about 10^{-43} seconds. If you're not familiar with the notation, 10^{-43} seconds is equal to one second divided by